Spatial Distribution of Bidirectional Charging Stations by Simulation and Evaluation of Traffic and Grid Data

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Abstract. This papers goal is the investigation on how bidirectional charging stations can be distributed across a dynamic area by combining traffic simulation with partially extracted grid data from official sources. We model charging demand as a spatiotemporal process and evaluate candidate locations with respect to accessibility, traffic flow, and waiting time. The proposed workflow integrates mobility traces, station utilization forecasts, and synthetic grid constraints to identify robust deployment strategies. Results indicate that jointly optimizing traffic and grid criteria can reduce peak grid load while improving charging availability in high-demand districts.

Keywords: Bidirectional charging, traffic simulation, SUMO, OpenStreetMap, spatio-temporal analysis, charging demand, V2G

Nomenclature

BEAM Behavior, Energy, Autonomy, and Mobility

EV Electric Vehicle

MATSim Multi-Agent Transport Simulation

POI Point of Interest

SoC State of Charge

SUMO Simulation of Urban Mobility

TraCI Traffic Control Interface

V2G Vehicle-to-Grid

OSM OpenStreetMap

CS Charging Station

1. Introduction

The rapid growth of electric mobility is shifting charging-infrastructure planning from static placement rules toward data-driven decision support [1]. Bidirectional charging adds a further dimension, since vehicles are no longer only energy consumers but can also feed power back to the grid, a concept known as Vehicle-to-Grid (V2G), which turns charging points into distributed flexibility assets [2]. Deciding where to place such stations therefore requires understanding where and when charging demand actually concentrates.

Because electric vehicles still represent a small share of the fleet, there is little real-world data on where charging stations should be placed and how intensively they are used. At the same time, uncoordinated charging concentrates electrical load in space and time and can stress local distribution grids [3], which makes the spatial and temporal structure of demand a central planning question. Existing siting studies, however, often rely on data or tooling that is not openly reproducible [4]. We therefore ask: *how can candidate locations for charging stations be ranked from openly available data, based on their accessibility and on how charging demand is distributed over space and time?*

To address this, we build a simulation workflow on openly available data. Road-network and land-use information are taken from OpenStreetMap (OSM), and traffic is simulated microscopically in Simulation of Urban Mobility (SUMO), controlled through its Python interface Traffic Control Interface (TraCI). From land-use Point of Interests (POIs)–residential buildings, offices, and amenities–we derive travel demand, and we model charging as a *spatiotemporal* process, that is, demand that varies simultaneously by location and time of day. The model distinguishes public charging stations from private home wallboxes, which are assigned to individual agents at their home location. A custom web interface, which is inspired by SUMO's web interface, lets users select an area of interest on a map and generate ready-to-run scenarios directly, lowering the barrier to reproducing and extending the analysis. Candidate sites are then assessed in terms of accessibility, station utilization, and the resulting local charging load.

The scope of this study is the planning and evaluation stage. We focus on the spatial and temporal distribution of charging demand and load under realistic mobility patterns. A detailed electrical model of the distribution grid (power flow, feeder and transformer constraints) as well as battery degradation and market or pricing mechanisms are beyond its scope and left to future work.

1.1 Motivation

Urban charging demand is highly uneven across time and location, which can cause queues at popular stations while other sites remain virtually unused. At the same time, cars that are parked for extended periods of time (in residential or commercial areas) can remain plugged in, creating opportunities for bidirectional energy exchange. The spatial distribution of charging stations therefore affects both user experience and grid load. For example, placing stations in high-traffic areas may improve accessibility but could also lead to congestion and increased waiting times. Conversely, locating stations in less busy areas might reduce queues but could limit their utilization and the potential for vehicle-to-grid (V2G) services. Understanding the spatial and temporal patterns of charging demand is therefore a prerequisite for infrastructure planning. However, there is a lack of open, reproducible methods for assessing candidate locations based on real-world mobility patterns and grid constraints.

1.2 Problem Statement

Given the afforementioned challenges, the problem statement of this paper can be formulated as follows: How can we determine the optimal spatial distribution of bidirectional charging stations, while regarding existing charging stations, in an urban area by integrating traffic simulation data with partially synthetic grid constraints, in order to maximize accessibility and minimize waiting times for electric vehicle users?

1.3 Contributions

This paper contributes a practical and reproducible workflow for planning bidirectional charging infrastructure with coupled traffic and grid evaluation. This work contains a user interface for deterministic scenario-based traffic generation along with a thorough analysis of candidate station portfolios under varying demand and grid conditions. By integrating synthetic grid constraints with mobility-based accessibility metrics, we provide a holistic framework for siting decisions that can be adapted to different urban contexts and policy objectives.

The main contributions of this work are as follows:

- A simulation-based siting pipeline that jointly evaluates mobility accessibility and distribution-grid impact.

- A multi-criteria scoring framework for comparing candidate station portfolios under proto-realistic demand variability.
- A reference implementation in Python to support reproducible experiments and scenario-based sensitivity studies.

2. Related Work

Planning charging infrastructure has been studied extensively, and recent reviews organize the field along the dimensions of station siting, sizing, grid reinforcement, trip-considerations and the treatment of uncertainty [5, 13]. Most contributions optimize either where demand is best served or how the grid copes with the additional load. Bidirectional charging, in which vehicles also feed power back, is only beginning to be integrated into such planning frameworks [9].

2.1 Traffic-Aware Siting

Traffic-aware siting methods estimate charging demand from origin-destination flows, activity patterns, and route choices, often placing stations so as to maximize coverage of traffic along routes. Data-driven analyses suggest that charging demand is largely shaped by proximity to amenities, Electric Vehicle (EV) density, and adjacency to high-traffic corridors, which motivates deriving demand from land-use data as we do [6]. Microscopic traffic simulators are increasingly used for this purpose. SUMO in particular provides built-in energy and charging-infrastructure models and has been used to assign electric vehicles to charging stations from simulated traffic data [7, 8].

A different approach than SUMO, is a mesoscopic traffic simulation model that uses a network of nodes and links to represent the road network, and simulates traffic flow based on traffic demand and road capacity. This approach can be used to estimate charging demand at a more aggregated level, and can be useful for planning charging infrastructure in areas with limited data availability or where detailed traffic simulation is not feasible. Multi-Agent Transport Simulation (MATSim) is an example of a mesoscopic traffic simulation model that has been used for this purpose, and paired with Behavior, Energy, Autonomy, and Mobility (BEAM) it is possible to simulate the impact of charging infrastructure on a dynamic traffic network [11, 12].

2.2 Grid-Aware Siting

Grid-aware planning approaches prioritize feeder hosting capacity, voltage stability, and peak-load mitigation, frequently relying on metaheuristic optimization to jointly site and size stations under electrical constraints [5]. While they improve technical feasibility, they can yield a poor user experience when travel behavior is not modeled in sufficient detail. The placement of charging stations is often motivated not only by charging demand and by economic advantages, but also by repercussions for the electrical network [14]. While a reasonable amount of studies have been conducted on the siting of charging stations, there are only few that consider working with large-scale eletric vehicle charging infrastructure [15].

2.3 Coupled and Bidirectional Approaches

A smaller but growing body of work couples transportation simulation with power-system analysis, for example through co-simulation frameworks that pass simulated charging loads to a distribution-grid model [10]. For bidirectional charging, planning models increasingly distinguish public and private chargers and account for uncertainty in demand and renewable generation [9]. Integrated and openly reproducible frameworks that combine microscopic mobility simulation with bidirectional charging at the level of individual public stations and private home wallboxes, however, remain limited. This paper addresses that gap on the mobility-and-demand side, while leaving a detailed electrical model of the distribution grid to future work.

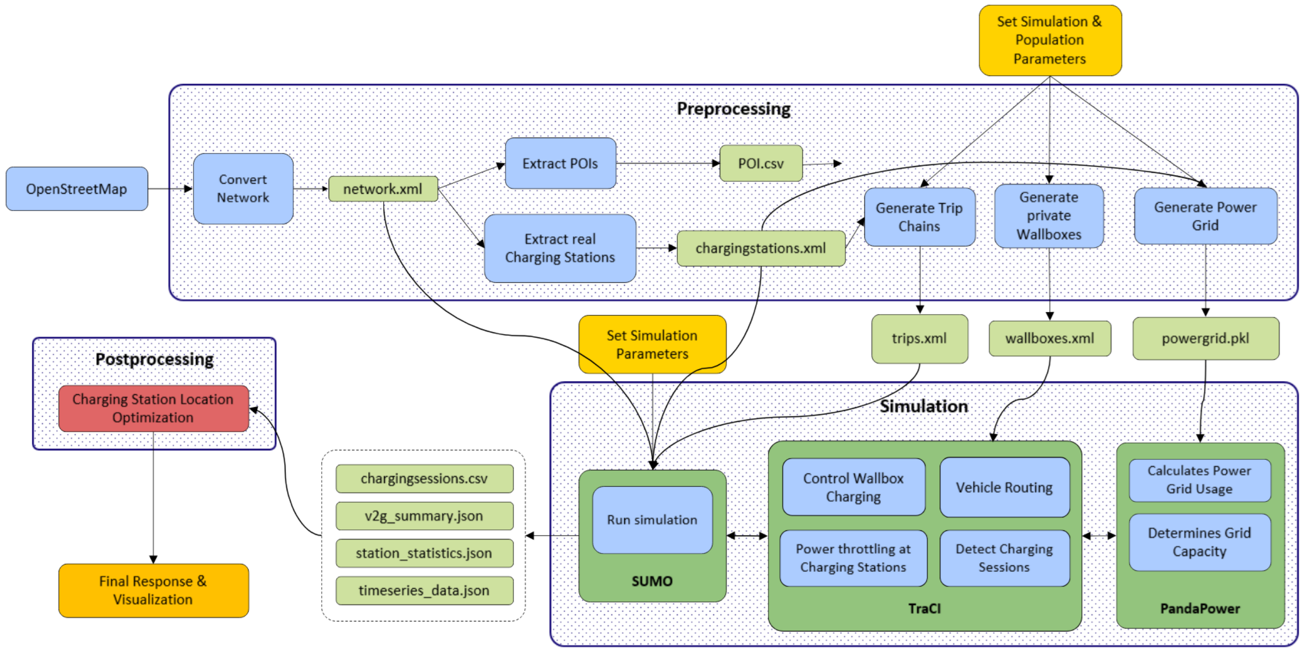

Figure 1: Complete Workflow Pipeline: From OSM data acquisition to scenario generation, execution and manipulation to analysis and visualization.

3. Methodology

SUMO is used as the microscopic traffic simulator and TraCI for real-time control, combined with a grid-impact model and a statistical distribution of home-charging devices, to evaluate the spatial distribution of bidirectional charging stations. Two siting strategies – *Saturate-and-Prune* and *Clustering* – are implemented and detailed in section 3.2.

3.1 System Overview

Figure 1 illustrates the complete workflow pipeline. The framework is organized into four consecutive stages: (i) data acquisition, (ii) preprocessing and scenario generation, (iii) coupled traffic–grid simulation, and (iv) analysis and visualization. Each stage produces intermediate artifacts that are consumed by the subsequent stage, so that the entire process is reproducible from a single set of user-defined parameters.

Data acquisition. The pipeline retrieves information like the road network, electrical grid topology, and candidate POIs from OSM. The road network is converted into a SUMO-compatible format using the `netconvert` tool, while grid features and POIs are persisted for later preprocessing. This ensures that the simulation has accurate and relevant data to work with, allowing for realistic modeling of traffic flow and charging behavior in the region of interest.

Preprocessing and scenario generation. The `chargingstations.xml` file contains both, existing and candidate stations produced by the placement algorithm, and is utilized not only for the generation of the synthetic power grid, but also in assistance with the POI file to generate trip chains. Trip chains map individual agents to a sequence of activities and destinations, from which their charging demand and low-State of Charge (SoC) behavior emerge. In parallel, the private-wallbox generator assigns home-charging devices to a share of agents following a statistical distribution, and the power-grid generator couples the charging stations and wallboxes to their nearest buses to form the synthetic distribution grid. To finalize the preprocessing, user-set parameters (EV penetration rate, wallbox share, battery capacities, grid voltage levels) are applied to the mechanisms of the trip chain, private wallbox, and power grid generation, yielding a fully specified scenario. Scenario generation is seeded and therefore deterministic; the coupled simulation is deterministic only once the interpreter's hash seed is fixed as well, as discussed in section 4.

Coupled traffic–grid simulation. After the preprocessing the files `osm.passenger.trips.xml`, `private_wallboxes.xml`, and `power_grid.pkl` will then be used by SUMO to simulate traffic flow and charging behavior. TraCI, SUMO and PandaPower are the key components of the simulation, which is executed in a time-stepped manner. TraCI allows for real-time control of the simulation, enabling the dynamic adjustment of traffic flow and charging behavior based on an Application Programming Interface (API) via Python. SUMO simulates traffic on a microscopic level (i.e., individual vehicles and their interactions), while PandaPower evaluates the impact of charging behavior on the synthetic distribution grid. While SUMO and TraCI handle the traffic simulation, SUMO and PandaPower do not communicate directly. Instead, TraCI acts as a bridge between the two, allowing for the exchange of information and control signals. This enables the simulation to capture the complex interactions between traffic flow and grid behavior, providing the ability for SUMO to incorporate grid constraints and for PandaPower to calculate the impact of charging behavior on the grid.

Analysis and visualization. The simulation produces a variety of output files, including `station_statistics.json`, `v2g_summary.json`, `charts_data.json`, `charging_sessions.csv` and `model_log_data.csv`. These persisted outputs are analyzed and visualized to evaluate the placement algorithm along metrics such as charging sessions, net energy flow over time, peak grid power, and individual charging station utilization (section 3.3). The results are rendered as plots and maps in the web

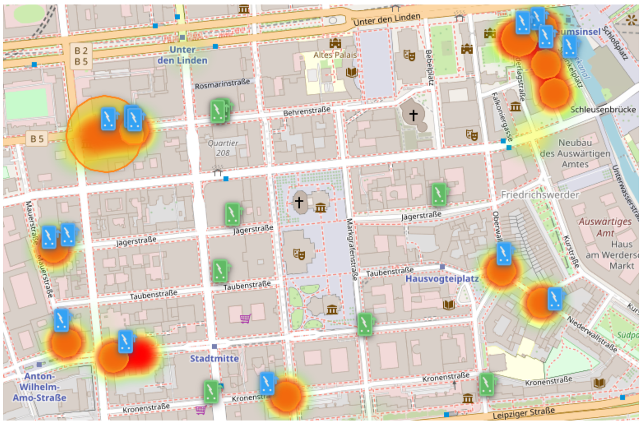

Figure 2: Resulting visualization with generated (blue) and extracted (green) CS for example run.

interface, and inform future iterations of the placement algorithm. Figure 2 shows an example visualization with generated (blue) and extracted (green) charging stations for a single run. The heatmap and orange circles indicate approximate areas for charging station placement, while the blue and green dots represent the actual locations of the generated and extracted charging stations, respectively.

3.2 Charging Station Placement Algorithm

Two siting strategies are implemented to compare different placement policies:

- **Saturate-and-Prune**: A naive iterative approach where candidate stations are placed on all viable road segments, ensuring maximum coverage. After each trial execution, charging stations with less than a 5% utilization rate are removed, and the simulation is rerun until all remaining stations exceed this threshold. A 5% threshold was chosen to ensure that stations are neither underutilized nor overburdened, while also allowing for a sufficient number of stations to be placed. This approach is designed to maximize acessibility and coverage, but may result in a higher number of stations being placed than necessary, leading to potential underutilization of some stations.
- **Clustering**: A heuristic that prioritizes locations with high traffic flow, potential

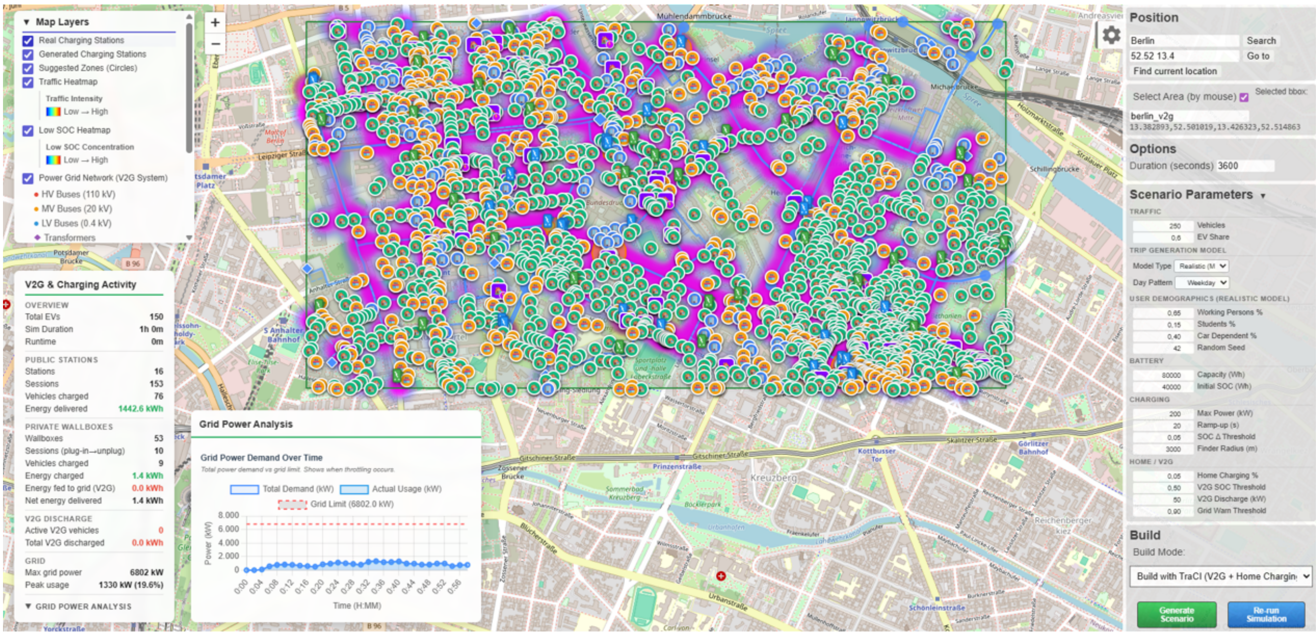

Figure 3: Example run of the workflow: the web interface with all data layers enabled.

traffic congestion, and consideration for vehicles that enter a state of low SoC. Placement follows the traffic flow, aiming to maximize accessibility where vehicles are most likely to need charging while reducing detours.

3.3 Evaluation Metrics

Mobility accessibility. CS-placement is evaluated based on traffic flow, charging behavior, energy consumption, and SoC. The gathered data is used to calculate the accessibility of the charging stations, which is defined as the percentage of agents that are able to access a charging station within a certain distance or time threshold. Distance and time are key factors in determining the accessibility of charging stations, as they directly impact the convenience and feasibility of using the stations for charging. Distance is an honest measure of how far an agent must travel to reach a charging station, while time takes into account factors such as traffic conditions, congestion, timely availability of charging stations, time and power (kWh) required to charge.

Grid impact. During the simulation, PandaPower solves an AC power flow on the synthetic distribution grid at every step, evaluating bus voltages and the loading of lines and transformers as constraints that bound the power allocated to each station. The grid impact reported per run is the aggregate demand and the peak grid power relative to the available grid capacity, which captures the stress the charging activity imposes on the grid without exceeding its voltage and loading limits.

Aggregation. The evaluation metrics are aggregated to provide a comprehensive assessment of the charging station placement algorithm. This includes combining mobility accessibility and grid impact metrics to evaluate the overall performance of the charging station network. The aggregation process involves analyzing the trade-offs between accessibility and grid impact, allowing for a balanced evaluation of the placement algorithm's effectiveness.

4. Experiments

4.1 Datasets

The experiment is conducted on mobility data, which is generated by an algorithm that simulates movement patterns of agents in a given area. The algorithm takes into account various factors such as land use, transportation infrastructure and natural behavior of agents to generate realistic mobility data. The generated data is then used to evaluate the performance of the proposed charging station placement algorithm. Before running the simulation, the dataset consists of surface level mobility (agent start- and endpoints, OSM-data), which is used for traffic flow, POIs and electrical grid information. When a simulation is run, the dataset is enriched with additional information such as charging behavior, energy consumption, specific locations of agents and their SoC. The datasets are generated using a combination of real-world data and synthetic data, ensuring that the results are both realistic and generalizable to other urban areas. The datasets are also designed to be scalable, allowing for the evaluation of the placement algorithms on larger areas with more agents and charging stations.

4.2 Experimental Setup

The reference scenario covers a one kilometer by one kilometer area in Berlin, Germany. It is populated with 250 agents whose home and activity locations are drawn from the residential, office and amenity POIs of the area. Each agent follows a daily trip chain: it departs between 6:30 and 9:00, stays at its activity location for eight hours, and returns home for the rest of the day. The simulation covers 24 hours of simulated time with a time step of 10 s, so the chains complete within the horizon and the evening and overnight period at home is part of the observation.

Of the 250 agents, 150 are electric vehicles with a battery capacity of 80 kWh, initialised at 50 % SoC; the remaining 100 are conventional vehicles. Every agent still carries a SUMO battery device, because the simulation is configured with `device.battery.probability` = 1, so the fleet-wide SoC statistics reported below cover all agents that entered the network, while charging is restricted to the 150 electric vehicles. Half of the electric vehicles are assigned a private home wallbox with a vehicle-specific type; 53 of those wallboxes could be placed on a home lane long enough to hold them, the remainder were skipped.

Charging power, V2G discharge and the wallbox logic are not part of the static SUMO scenario. They are injected at runtime through TraCI, which overrides the station power, applies the ramp-up, and arbitrates the total power draw against the synthetic distribution grid through a PandaPower model of 37 buses, 88 lines and 15 transformers. Charging power per station is capped at 200 kW with a 20 s ramp-up, V2G discharge is enabled above 50 % SoC with up to 50 kW per vehicle and is triggered when the grid reaches 90 % of its capacity. The complete parameter set is persisted per run in `sim_params.json`, so that every reported figure can be traced back to the configuration that produced it.

The public infrastructure is the one already present in the area, that is, the 26 charging stations extracted from OSM. One of those stations sits on a 0.22 m lane and was written with an extent of 0.01 m, which SUMO rejects. Because the rejection aborts parsing of the remaining additional file, the twelve stations declared after it were dropped as well, leaving only 14 of the 26 stations in the network. The station extent was corrected before the run reported here, so that all 26 stations are present. The run quantifies how the existing infrastructure serves the fleet, and it provides the charging-demand log from which the Clustering strategy derives its candidate sites.

Reproducibility. Repeated executions of an identical configuration were found to differ. Three unseeded runs delivered between 372.7 and 399.2 kWh of public energy in 48 to 52 sessions, and between 1600.8 and 1751.0 kWh of wallbox energy in 52 to 59 sessions, a spread of roughly 7 % and 9 % respectively. The cause is the iteration order over the set of active electric vehicles, which determines the order in which vehicles request power from the shared pool and which varies between processes because Python randomises string hashing. Fixing the hash seed makes the run bit-identical; the figures reported below come from such a fixed-seed execution. Nevertheless, a single run cannot separate systematic effects from the noise that dynamic routing introduces, so the numbers are reported as observations of one scenario instance rather than as statistically significant differences between the two placement strategies. A multi-seed comparison of Saturate-and-Prune against Clustering along the metrics of section 3.3 is the subject of ongoing work and is discussed in section 5.

4.3 Results

Where the energy is drawn. Over the simulated day the fleet draws 2127.9 kWh in 109 charging sessions, and the split between the two kinds of infrastructure is the central observation. Private wallboxes deliver 1737.6 kWh in 59 sessions to 41 vehicles, while the 26 public stations deliver 390.2 kWh in 50 sessions to 29 vehicles. Home charging therefore accounts for 82 % of the energy at a comparable number of sessions, because a wallbox session lasts 3.9 hours on average whereas a public session ends when the agent resumes its trip. Only 13 of the 26 public stations record a session at all and only 8 of them deliver energy. A single station, `real_cs_12095948807`, carries 15 of the 50 public sessions and 37 % of the public energy. This is the placement problem the siting strategies address: the public network is not short of stations in aggregate, but the few stations that lie on the routes the agents actually drive absorb the demand while the rest stay idle.

Accessibility. Despite the available home charging, the fleet ends the day below its starting point. The mean SoC drops from 50.0 % to 33.2 % with a wide spread (std. 20.2 %, median 46.2 %). Of all agents, 31.4 % end below 20 % SoC, 23.3 % drop below 10 % at some point during the day, and 4.5 % are fully depleted. The gap between mean and median indicates that the distribution is pulled down by a tail of agents rather than by a uniform decline, and that tail consists of the vehicles without a wallbox, which depend on the eight public stations that actually deliver. Table 1 summarises the run.

Table 1: Baseline run on the existing OSM infrastructure: 250 agents, 24 hours of simulated time, 10 s time step, coupled through TraCI.

Metric	Value
Agents	250 (150 EV)
Logged vehicle states	1 203 472
Simulation horizon	24 h (8640 steps)
Mean SoC, start → end	50.0 → 33.2 %
Std. of end SoC	20.2 %
Agents below 20 % SoC at end	31.4 %
Agents below 10 % SoC (any time)	23.3 %
Agents fully depleted	4.5 %
Public CS available (OSM)	26
Public CS delivering energy	8
Public sessions / vehicles	50 / 29
Public energy	390.2 kWh
Private wallboxes placed	53
Wallbox sessions / vehicles	59 / 41
Wallbox energy	1737.6 kWh
V2G energy discharged	0.07 kWh
Peak grid draw	601.4 kW
Low-SoC samples in clusters	65 117
Demand clusters retained	57 (of 110)
Distinct vehicle visits	77
Energy demand of clusters	3809 kWh
Charge points estimated	58

Bidirectional operation. V2G discharge is enabled but contributes 0.07 kWh over the whole day. The grid reaches a peak draw of 601.4 kW, and the discharge condition of 90 % grid capacity is almost never met, so the bidirectional capability of the fleet stays unused in this scenario. The run therefore characterises charging demand; it does not yet demonstrate a V2G benefit, which would require either a substantially higher EV penetration or a distribution grid dimensioned closer to the load.

Derived charging demand. The DBSCAN stage of the Clustering strategy operates on the samples in which a vehicle is below 30 % SoC, of which the run logs 221 349; half of them are drawn at random to bound the clustering cost. The stage returns 110 clusters, of which 53 fall within 100 m of an already existing station and are discarded, leaving 57 candidate sites that carry 65 117 low-SoC samples. A sample is one vehicle at one 10 s time step, so the sample count measures dwell time rather than demand, and a vehicle parked at home overnight contributes thousands of samples without ever asking for a public charge. The demand of a cluster is therefore derived from the vehicles themselves. Samples inside a cluster are grouped per vehicle and split into separate visits whenever a vehicle is absent for more than five minutes, and each visit is charged with the energy it would need to reach 80 % SoC. Across all clusters this yields 77 visits and an energy demand of 3809 kWh. Table 2 lists the ten clusters with the highest demand. Demand is dispersed rather than concentrated: the two largest clusters carry 21 % of the visits and 21 % of the energy demand, and 49 of the 57 clusters consist of a single visit by a single vehicle. The mean SoC inside a cluster ranges from 0.0 % to 29.8 % (mean 18.9 %), which separates clusters that mark genuine range emergencies from those that merely mark long parking. Cluster radii stay compact at 25.0 m to 49.0 m (mean 25.8 m), so each cluster is small enough to be served by a single station site.

Table 2: The ten charging-demand clusters of the baseline run with the highest energy demand. *Vis.* is the number of distinct vehicle visits, *Veh.* the number of vehicles behind them, *kWh* the energy those visits need to reach 80 % SoC, *CP* the resulting number of charge points and *Cap.* the grid capacity at the nearest connection point.

ID	Vis. (#)	Veh. (#)	SoC (%)	Rad. (m)	kWh	CP (#)	Cap. (kW)
80	14	13	16.6	49.0	703.4	2	5000
92	2	2	18.4	25.0	102.8	1	5000
85	2	2	27.9	45.2	102.2	1	500
98	2	2	16.6	25.0	102.0	1	5000
81	2	2	18.6	25.0	98.1	1	500
8	2	2	19.5	25.0	96.2	1	5000
38	2	2	26.9	25.0	85.3	1	500
109	2	2	27.8	25.0	85.0	1	500
95	1	1	0.0	25.0	64.0	1	5000
58	1	1	4.0	25.0	60.8	1	5000

All 57 clusters: 77 visits, 3808.9 kWh, 58 charge points; SoC 18.9 ± 7.4 %; radius 25.8 ± 4.1 m; grid rating 25 excellent, 32 good.

Grid-aware siting. All 57 clusters fall within reach of the synthetic distribution grid, at a mean distance of 112 m to the nearest connection point (max. 272 m). Twenty-five clusters are rated *excellent* and thirty-two *good* with respect to available grid capacity, and none had to be discarded for lack of a feasible connection. Seventeen clusters are limited to a 500 kW connection while the remaining forty have 5000 kW available. A charge point is a single connector at which one vehicle can charge, so a station site aggregates several of them; the heuristic sizes a cluster as its hourly energy demand divided by the throughput of one 50 kW charge point at 75 % utilisation, and then caps the result at what the local grid connection can supply. Spreading the demand over a 17.5-hour observation window lowers the hourly rate so far that all but one cluster are sized at a single charge point, and the grid cap never binds. The grid connection is therefore not the limiting factor in this scenario, but the sizing rule itself is too coarse: dividing daily energy by the length of the observation window averages away the evening peak that a station actually has to serve.

Interpretation. The run supports the premise of the approach on the public side and qualifies it on the private side. Only 8 of 26 public stations deliver energy along the routes the agents drive, and a single station absorbs 37 % of the public energy, which is the misplacement that a traffic- and SoC-driven heuristic is meant to correct. At the same time, private wallboxes cover 82 % of the delivered energy, so the marginal value of additional public stations is limited to the agents without home charging, and a siting study that ignores private infrastructure would overstate the need for public stations. Over the full day, demand does not collapse onto a handful of sites: the two largest clusters account for a fifth of the visits, and 49 of 57 clusters represent a single vehicle stopping once. A clustering heuristic is therefore expected to gain less over uniform saturation than a shorter observation window suggests. Deciding between the two strategies requires the multi-seed comparison described in section 5. The numbers reported here establish the baseline against which it will be measured.

5. Discussion

5.1 Practical Example

5.2 Limitations

Wallbox access control. Private wallboxes are declared with a vehicle-type restriction so that only the owning agent may use them. In the reported run this restriction is not enforced: 30 of the 53 wallboxes delivered energy to vehicles other than their owner, and across the fleet foreign vehicles drew 972.8 kWh against 63.0 kWh drawn by the owners themselves. The wallboxes are visible to SUMO through the additional file and are charged by any vehicle that stops on the corresponding lane, while the access rule is only honoured by the TraCI layer that manages them as virtual stations. The split between public and private energy reported in section 4.3 should therefore be read as a split between kerbside and home-location charging, not as evidence that private access control works. Enforcing the restriction requires removing the wallboxes from the static scenario so that they exist only inside TraCI; this is implemented in a separate development branch that does not yet carry the remaining functionality, and merging the two is left to future work.

Bidirectional operation. Although V2G discharge is implemented and enabled, it contributes 0.07 kWh over the simulated day. The discharge condition requires the grid to reach 90 % of its configured capacity, which a fleet of 150 electric vehicles in a one square kilometer area does not achieve against the synthetic grid. The scenario therefore demonstrates the mechanism but not its effect, and any statement about the value of bidirectional charging would require a higher EV penetration or a distribution grid dimensioned closer to the load.

Single-run evidence. All figures come from one scenario instance. The comparison between Saturate-and-Prune and Clustering is not yet supported by the data, because a single run cannot separate the effect of a placement strategy from the variation that dynamic routing introduces. A multi-seed study across both strategies is the immediate next step.

Charge-point sizing. The heuristic sizes a cluster by dividing its energy demand by the length of the observation window. Over a 24-hour horizon this averages away the evening peak at which home and kerbside charging coincide, and consequently sizes almost every cluster at a single charge point. A peak-based rule, applied to the busiest hour rather than to the daily mean, would give a more useful estimate.

Grid model. Grid capacity at a candidate location is derived from the voltage level of the nearest bus and the load already connected to it, not from a power-flow solution that evaluates line loading and voltage limits along the feeder. The reported grid ratings therefore indicate proximity to a connection point of sufficient nominal capacity rather than confirmed hosting capacity.

6. Conclusion

This paper presented a simulation-driven method for distributing bidirectional charging stations under joint traffic and grid criteria. The evaluation demonstrates that coordinated planning can improve both charging accessibility and electrical reliability.

6.1 Future Work

Future work will include stochastic optimization under forecast uncertainty and integration of dynamic pricing signals. We also plan to validate the framework with larger real-world datasets and pilot deployments in mixed urban-suburban regions.

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References

[1] International Energy Agency (IEA), *Global EV Outlook 2023: Catching up with Climate Ambitions*. Paris, France: IEA, 2023. [Online]. Available: https://www.iea.org/reports/global-ev-outlook-2023

[2] W. Kempton and J. Tomić, "Vehicle-to-grid power fundamentals: Calculating capacity and net revenue," *Journal of Power Sources*, vol. 144, no. 1, pp. 268–279, 2005.

[3] K. Clement-Nyns, E. Haesen, and J. Driesen, "The impact of charging plug-in hybrid electric vehicles on a residential distribution grid," *IEEE Transactions on Power Systems*, vol. 25, no. 1, pp. 371–380, 2010.

[4] F. He, D. Wu, Y. Yin, and Y. Guan, "Optimal deployment of public charging stations for plug-in hybrid electric vehicles," *Transportation Research Part B: Methodological*, vol. 47, pp. 87–101, 2013.

[5] "A review on electric vehicle charging station planning: Infrastructure placement, sizing, upgrades, and uncertainties," *Journal of Energy Storage*, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2352152X25040381

[6] "Data-driven optimization of EV charging station placement using causal discovery," *arXiv preprint arXiv:2503.17055*, 2025. [Online]. Available: https://arxiv.org/abs/2503.17055

[7] T. Kurczveil, P. Á. López, and E. Schnieder, "Implementation of an energy model and a charging infrastructure in SUMO," in *Simulation of Urban Mobility (SUMO)*, https://www.researchgate.net/publication/276063158_Implementation_of_an_Energy_Model_and_a_Charging_Infrastructure_in_SUMO,

[8] "Assignment approach for electric vehicle charging using traffic data collected by SUMO," *World Electric Vehicle Journal*, vol. 14, no. 2, art. 40, 2023. [Online]. Available: https://www.mdpi.com/2032-6653/14/2/40

[9] "Vehicle-to-grid enabled charging infrastructure planning and operations considering demand uncertainties," *Transportation Research Part D: Transport and Environment*, 2023. [Online]. Available: https://www.sciencedirect.com/science/article/abs/pii/S1361920923003152

[10] "Assessing EV charging impacts on power distribution systems: A unified co-simulation framework," *arXiv preprint arXiv:2505.21773*, 2025. [Online]. Available: https://arxiv.org/abs/2505.21773

[11] A. Horni, K. Nagel, and K. W. Axhausen, Eds., *The Multi-Agent Transport Simulation MATSim*. London, U.K.: Ubiquity Press, 2016. [Online]. Available: https://doi.org/10.5334/baw

[12] C. Sheppard, R. Waraich, A. Campbell, A. Gopal, and A. Pozdnukhov, "Modeling plug-in electric vehicle charging demand with BEAM: The framework for behavior, energy, autonomy, and mobility," Lawrence Berkeley National Laboratory, Berkeley, CA, USA, Tech. Rep. LBNL-2001018, 2017. [Online]. Available: https://www.osti.gov/biblio/1398472

[13] R. Pagany, L. Ramirez Camargo, and W. Dorner, "A review of spatial localization methodologies for the electric vehicle charging infrastructure," *International Journal of Sustainable Transportation*, vol. 13, no. 6, pp. 433–449, 2019. [Online]. Available: https://doi.org/10.1080/15568318.2018.1481243

[14] F. Ahmad, A. Iqbal, I. Ashraf, M. Marzband, and I. Khan, "Optimal location of electric vehicle charging station and its impact on distribution network: A review," *Energy Reports*, vol. 8, pp. 2314–2333, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2352484722001809

[15] "Large-scale optimization of electric vehicle charging infrastructure," in *Proc. 32nd ACM Int. Conf. on Advances in Geographic Information Systems (SIGSPATIAL '24)*, 2024. [Online]. Available: https://dl.acm.org/doi/10.1145/3678717.3700830

[16] V. Lazari and A. Chassiakos, "Multi-objective optimization of electric vehicle charging station deployment using genetic algorithms," *Applied Sciences*, vol. 13, no. 8, art. 4867, 2023. [Online]. Available: https://www.mdpi.com/2076-3417/13/8/4867